

Section 12.3: The Dot Product

Vectors and the Geometry of Space

MTH 310
Calculus III

The Dot Product

Dot Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Multiply corresponding components and add.

Computing Dot Products

Example 1

Compute each dot product.

$$\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2 \text{ — the dot product of a vector with itself gives the square of its magnitude.}$$

Properties of the Dot Product

Properties

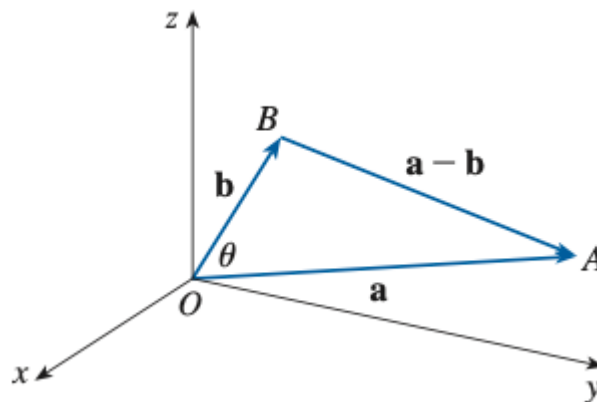
1. $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
2. $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (commutative)
3. $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ (distributive)
4. $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$
5. $\vec{0} \cdot \vec{a} = 0$

The Geometric Meaning — Angles

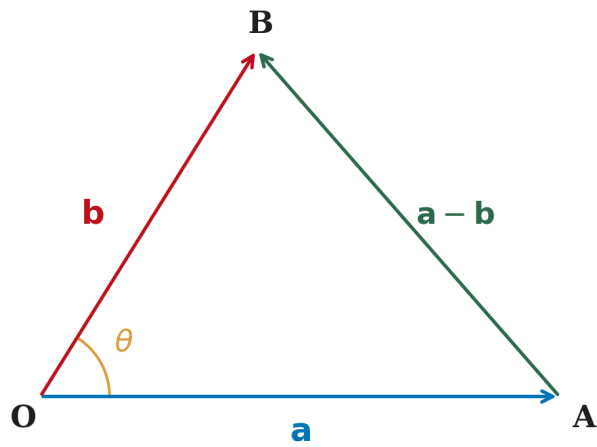
Geometric Form of the Dot Product

If θ is the angle between $\vec{a}$ and $\vec{b}$ (with $0 \leq \theta \leq \pi$), then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$



Example: Finding the Angle Between Vectors

Example 2

Find the angle between $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$.

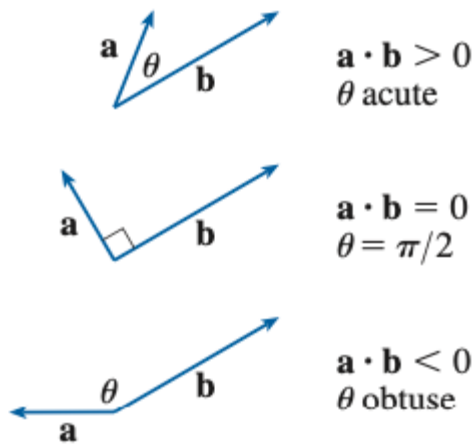
Compute a dot b, then $|\vec{a}|$ and $|\vec{b}|$, then use the angle formula.

Orthogonality

Orthogonal Vectors

Two vectors $\vec{a}$ and $\vec{b}$ are **orthogonal** (perpendicular) if and only if

$$\vec{a} \cdot \vec{b} = 0$$



This is the **most important** use of the dot product — a quick test for perpendicularity!

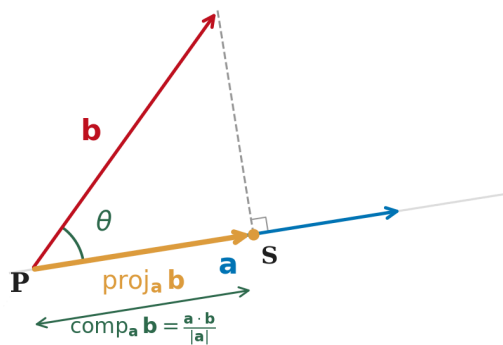
Your Turn — Orthogonality Check

Example 3

Are $\vec{u} = \langle 2, 1, -1 \rangle$ and $\vec{v} = \langle 1, -1, 1 \rangle$ orthogonal?

Compute $u \cdot v$ and check if it equals zero.

Projections



Scalar and Vector Projections

Scalar projection (component of $\vec{b}$ onto $\vec{a}$):

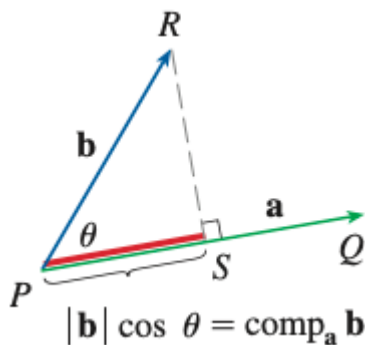
$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

This is a **number** — the signed length of the shadow.

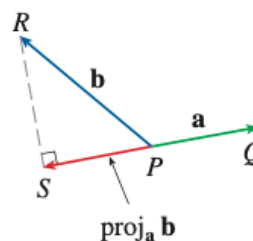
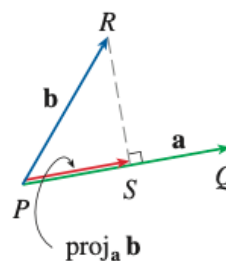
Vector projection of $\vec{b}$ onto $\vec{a}$:

$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

This is a **vector** — the actual shadow vector along $\vec{a}$.



comp = how far, \text{space proj} = the actual vector



Example: Computing Projections

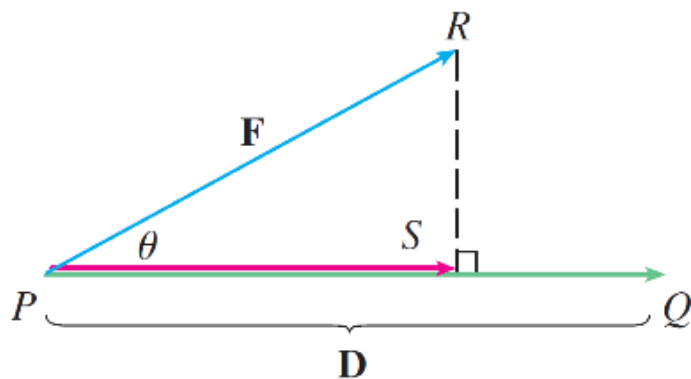
Example 4

Find the scalar and vector projections of $\vec{b} = \langle 12, 1, 2 \rangle$ onto $\vec{a} = \langle -1, 4, 8 \rangle$.

Compute $\vec{a} \cdot \vec{b}$, then $|\vec{a}|$, then use the projection formulas.

Work — A Physical Application

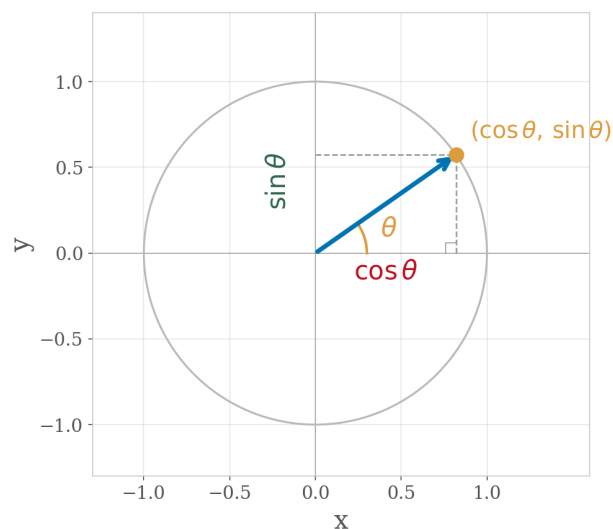
$$W = \text{comp}_{\vec{D}} \vec{F} \cdot |\vec{D}| = \vec{F} \cdot \vec{D}$$



Example 5

A wagon is pulled 100 m by a force of 70 N applied at 35° above horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.



$$W = \vec{F} \cdot \vec{D} = 70 \cos 35^\circ \cdot 100 + 70 \sin 35^\circ \cdot 0 = 7000 \cos 35^\circ \approx 5734 \text{ N} \cdot \text{m}$$

Your Turn — Work

Example 6

A box is pushed 5 m along the floor by a 40 N force applied at 20° below horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.

Homework

Section 12.3: 5, 7, 11, 19, 23ab, 27, 39, 41, 45, 49, 57